

Isekai Strategem — Complete Game Design Document

Status: Living document, current as of `main`. This is the single authoritative design reference: it explains every system, the numbers behind it, and the full content catalogs. It consolidates and supersedes the older, partial docs (`combat_v2_design.md` is the rework spec / build tracker; the `packages/battle_engine/doc/*` reference files predate combat v2 and are stale).

Inspirations: World Trigger (Trion, Triggers, Black Triggers, the Attacker/Shooter/Sniper/Trapper classes), Naruto (chakra-style resource play, genjutsu/shadow/seal status flavor), and tabletop D&D (the d20 to-hit engine, advantage/disadvantage, critical hits, conditions).

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1. What the game is

Isekai Strategem is a **turn-based 3v3 tactical squad battler**. Each player fields a **squad of three characters**, each kitted with a **Loadout** of Triggers (and optionally a Black Trigger). Two squads alternate turns, committing

actions that resolve through a fixed phase order, until one squad is wiped. A single character can be deleted quickly, so the game rewards planning around resource tempo, status setups, counters, and squad synergy rather than raw stat-checking.

There is also a story/visual-novel mode scaffold; this document covers the combat game.

Two layers of code:

- `packages/battle_engine/` — pure-Dart engine: combat resolution, counters, uniques, status effects, catalogs (triggers / black triggers / roster), AI.
- `app/` — Flutter app: the queue/resolution orchestration (`play_session.dart`), Team Efficiency Grade, draft/loadout/target selection, screens, and the Supabase accounts/XP integration.

2. The gameplay loop

- Draft** three characters into a squad (each has a fixed base stat block, a type, and an innate perk).
- Build a Loadout** for each: equip Triggers within that character's Trion Capacity budget, satisfying the loadout rules (below).
- The engine computes each squad's **Team Efficiency Grade (TEG)** from the drafted loadouts — a build-quality score that grants dice advantages in battle and inversely scales post-battle XP.
- Battle**: squads alternate turns. On your turn you queue one or more actions (more if FAT triggers), choose targets, and end the turn; the queue resolves through the six-phase order, then the opponent responds.
- Resolve to a winner**; award inverse-TEG XP to the player's account.

3. Squads, characters, and character types

A squad is **exactly three characters**. The roster has **20 characters** across four **types** (see Appendix B for the full stat table):

Type	Count	Role identity	Stat signature
Attack	5	Front-line damage	High Attack (22–27), low Defense, low Team Spirit (offense pole)
Defense	5	Damage soak / control	High Defense (14–16) & Armor, mid Team Spirit
Support	5	Heals / buffs / Trion	High Trion Affinity & Team Spirit (sustain pole), low Attack
Unique	5	Gimmick specialists	Built around a signature Unique ability + a distinctive perk

Character type also feeds the TEG's Team Spirit Alignment and Stat Coherence sub-scores (a squad whose stats and loadouts match their pole scores higher).

4. Stats

Every character has a base **Stats** block. All stats are visible in the UI (the loadout/roster panels surface even the "hidden" ones).

Stat	What it does
Attack	Added to the attacker's d20 to-hit roll; the dominant contributor to landing hits.
Defense	Added to the defender's d20 roll (opposed); the target's evasion/guard.
Armor	Flat damage subtracted after multipliers, every hit.

Stat	What it does
Max Health	Hit points (baseline 100 across the roster). At 0, the character is DEFEATED.
Trion Capacity	The character's Trion pool size / Loadout equip budget (see §6, §7). ~100–130.
Trion Affinity	Raises the chance of higher per-turn Trion income tiers (§6).
Team Spirit	Dual-direction stat: below 50 boosts offense, above 50 boosts sustain (§5).
Critical Chance	Chance an attack is a critical hit (widens the crit die-threshold; §11).
FAT Chance	Chance Full Arms Trigger activates on a turn, unlocking multi-action turns (§8).
Status Infliction	The "save DC" side of applying status effects.
Status Resistance	The "saving throw" side of resisting inflicted statuses.

5. Team Spirit (dual-direction stat)

Team Spirit runs **0–100 with 50 neutral**. It is two independent linear scalers from the midpoint out to each extreme (`TeamSpiritCurve`):

- **Below 50 → offense pole.** Scales up to, at Team Spirit 0:
 - +30% single-target damage, +30% burst damage, +20 flat Critical Chance.
- **Above 50 → sustain pole.** Scales up to, at Team Spirit 100:
 - +30% healing received, +20 flat FAT Chance.
- **At 50:** neutral, no bonus either way.

So an attack character (Team Spirit ~30–40) is a committed glass cannon, while a support (~60–70) is a committed sustainer. The UI shows the live offense/sustain readout next to the value. This stat is a major TEG input (Team Spirit Alignment rewards a squad whose loadouts match their pole).

6. Trion (the resource economy)

Trion is the resource that pays for abilities. It exists as two related quantities (per the World Trigger "Trion capacity" concept):

- **Trion Capacity** — the size of the pool and the **Loadout equip budget** (a Trigger has an *equip cost* paid against Capacity at build time, and a *Trion cost* paid from the running pool at use time).
- **Per-turn Trion income** — at the start of a turn each side rolls a tier:

Tier	Gain	How it's rolled
Low	+10	Base state
Medium	+20	$35\% + 1\% \cdot \Sigma \text{TrionAffinity}$ chance to upgrade Low → Medium
High	+35	If Medium reached, a further $20\% + 1\% \cdot \Sigma \text{TrionAffinity}$ to upgrade

Trion Affinity therefore makes higher income more likely; a high-Affinity support like Haru Ellison (Affinity 32) is the squad's battery.

Trion denial is a real axis: `sapped` drains 25% of Trion Capacity, `genjutsu_trapped` drains 15%, `silenced` blocks abilities, `overcharged` halves costs, `choked` doubles them.

Design note (see reviews): income is currently a semi-random tier and most turns spend one ability, so Trion capacity rarely bites. A future pass may make income deterministic and capacity-gate FAT.

7. Triggers and Loadouts

Triggers are the equippable abilities. A **Loadout** is a character's set of equipped Triggers plus an optional Black Trigger.

Loadout rules (`LoadoutRulesConfig`):

- **Max 8 equipped items** total (Triggers + the Black Trigger count together).
- **Exactly 4 active abilities** required (an `ActiveTrigger` is 1; a Black Trigger contributes however many active abilities it carries).
- **Total equip cost $\leq$ the character's Trion Capacity.**

So loadout-building is a knapsack: fit exactly four usable abilities plus passive support Triggers under the Trion budget. Passive Triggers (counters, stat passives) fill the other slots.

Trigger anatomy (fields that define an `ActiveTrigger`): id, name, **category** (functional role), equip cost, Trion cost, cooldown (turns), origin tag, range tag (melee/ranged), attack type, **attack subtype** (single / burst / aoe), damage type, damage (a `DiceExpression`, `NdM+flat`), target count, hits per use, optional heal, inflicted status effects, optional **unique behavior** or **reactive kind**.

Trigger categories and the catalog split (60 active Triggers total — see Appendix A):

Category	Count	Role (World Trigger analogue)
attacker	16	Melee bruisers (Attacker: Kogetsu/Scorpion)
shooter	9	Volume fire / bursts (Gunner/Shooter: Asteroid/Viper)
sniper	3	Long-range big hits (Sniper: Ibis/Lightning)
trapper	24	Control, DoT, debuffs, most uniques (Trapper: Spider/Escudo)
optional	20	Bufs, wards, self/utility, reactive counters

(Categories overlap the 60; some Triggers count in the flavour role that best fits. The engine balance target is a broad, distinct spread — see §12/Appendix A.)

8. Full Arms Trigger (FAT)

Normally a character uses **one ability per turn**. **FAT** (`FatConfig`) is the burst valve:

- Each turn, a character can **trigger FAT** (chance = its FAT Chance stat, plus Team Spirit's sustain-pole bonus).
- On a FAT turn, that character may use **up to 3 abilities**.
- **Cost of greed:** if 2+ abilities are used on the FAT turn, a penalty applies — **cooldowns are doubled** and **Trion Affinity is halved** — and FAT then **locks out for 3 turns**.

FAT is the coordinated-nuke lever, deliberately taxed so it's a tempo decision, not free. The **Draegor** passive counter specifically punishes an *opponent* who FAT-chains 2+ abilities.

9. The turn & queue model

Turns **alternate** between squads. Cross-team who-goes-first (initiative) is currently an **even 50/50 coin flip** at battle start. A 15-second turn timer exists (`secondsPerTurn` = 15).

On the player's turn:

1. **Queue** actions (character + ability + targets). Trion is **spent at queue time**; **un-queueing refunds** it. Cooldowns are applied at resolve time.
2. Ability legality is enforced live (affordability, cooldown, targeting, the per-turn ability limit which scales with FAT).
3. **End Turn** → the queued actions **resolve** through the six-phase order (§10), results are shown after a brief cosmetic "Resolving..." beat, then the AI opponent takes its turn through the same path.

The AI builds and resolves a full queue through the *same* resolution path as the player (important for future server authority).

10. Resolution order (the six phases)

Within a turn, queued actions do **not** resolve in the order queued. They resolve by a fixed **phase order**, then by a within-phase tiebreak. The phases group actions by kind so that setup lands before payoff:

1. Pre-emptive / reactive arming
2. **Bufs** (self/ally empowerment, wards)
3. **Control** (no-damage debuffs / crowd control)
4. **Attacks** (anything that deals damage)
5. Post-attack effects
6. Cleanup / expiry ticks

Within a phase, ties break by **Team Spirit deviation from 50** (the more committed character acts first), and finally by queue order. Example: a buff resolves before an attack queued earlier; between two attacks, the higher Team-Spirit-deviation character strikes first.

Reactive expiry (timed wards/traps/marks) is ticked once per turn so temporary effects time out.

11. Combat resolution math (to-hit, crit, damage)

Every damaging hit resolves in two independent rolls: an **opposed d20 to-hit**, then a **separate damage roll**.

To-hit (opposed d20):

- Attacker rolls `d20 + Attack`; defender rolls `d20 + Defense`. Attacker total $\geq$ defender total → **hit**, else **miss**.
- **Advantage / disadvantage** roll two dice and keep the best / worst (perks and statuses grant these — e.g., Airi's *Feint* gives the first attacker against her disadvantage).

Critical hits:

- A hit is a **critical hit** if the attacker's natural d20 lands in the crit range. Crit chance maps to a widening threshold: at 0% crit chance you crit only on a natural 20 (`thresholdAtMinChance = 20`); at max crit chance the threshold drops to 5 (`thresholdAtMaxChance = 5`), i.e. crit on a natural 5+.
- A crit **doubles damage** (`criticalHitDamageMultiplier = 2.0`).
- A **critical miss** (natural low) penalizes the attacker (−20% Defense and −20% Team Spirit for 1 turn).

Damage formula (the log shows every step):

```
diceTotal          = the Trigger's own damage roll (NdM + flat)
× preCritMultiplier = Team Spirit / perk / status outgoing-damage multiplier
× crit (×2 if crit)
− Armor            = the target's flat Armor
```

$\times$ resistance/vuln = damage-type resistance ($\times 0.5$) or status multiplier
= final damage

Burst abilities roll this per hit; AoE applies to each target.

Design note (see reviews): because flat stat modifiers (Attack ~ 26 , damage flat $+37\dots+86$) dwarf the d20's ± 19 swing, the roll currently contributes $\sim 15\%$ of outcomes — the "no bounded accuracy" issue flagged for a P0 rebalance.

12. Status effects

There are **~ 60 status effects** (`StatusEffectCatalog`), each with a tuned duration and magnitude (`StatusEffectMagnitudes`; full values in Appendix C). They fall into families:

- **Hard control:** stunned (1t), frozen (1t), prone (1t), charmed (3t), petrified (2t, but also halves damage taken), silenced (1t).
- **Damage-over-time:** bleeding (8/t), scorched (12/t), necrotic_wound (12/t), poisoned, corroded (also -3 Armor).
- **Damage-taken multipliers:** marked ($\times 1.5$, 1t), exposed ($\times 1.25$, 2t), guarded ($\times 0.75$), petrified ($\times 0.5$).
- **Outgoing-damage swings:** empowered ($\times 1.25$), enraged ($\times 1.5$ but -3 Def), weakened ($\times 0.75$).
- **Stat swings:** inspired ($+2/+2$), fatigued ($-2/-2$), hastened, slowed, chilled, scorched, shattered_guard, braced, focused, rallied ($+20$ max HP).
- **Economy / disruption:** sapped (-25% Trion cap), genjutsu_trapped (-15%), overcharged ($\times 0.5$ cost), choked ($\times 2$ cost), suppressed (-5 infliction), warded ($+10$ resistance), hexed (-10 resistance).
- **Utility / signature:** regenerating, radiant_blessing, adrenaline_rush ($+15$ crit), battle_trance ($+20$ FAT), misfire (50% self-fail), interdict, forced_repetition, isolation, untargetable, called_shot, minds_eye_reveal, etc.

Applying a status is contested: **Status Infliction (attacker) vs Status Resistance (target)**, with immunities bypassing the roll.

Design note: statuses are not yet on a shared power budget; magnitudes/durations vary widely (marked $\times 1.5/1t$ vs exposed $\times 1.25/2t$; DoTs 8 vs 12), a flagged tuning target.

13. Counters (reactive + passive)

Counters are the reaction economy — armed on your turn, they fire when the opponent acts into them. There are **two kinds**.

13.1 Reactive counters (armed → fire on the opponent's action)

Driven by `ReactiveKind`:

Kind	Effect
reflectNonAoe	Reflect the next non-AoE attack back at the attacker.
dodgeMeleeSingle	Dodge the next single-target melee attack (e.g., <code>predictive_parry</code>).
negateByOrigin	Negate the next attack of a matching origin type.
burstMitigation	Reduce incoming burst damage (<code>numbing_toxin</code>).
cooldownSabotage	Extend an attacker's cooldowns when they act (<code>frozen_tempo</code>).

Kind	Effect
<code>redirectToOwnAlly</code>	Redirect an incoming attack onto one of the attacker's allies.
<code>trapOnAction</code>	Spring a trap when the target takes an action.
<code>nullifyAoe</code>	Nullify an incoming AoE (<code>death_ledger</code> — engine signals the app to borrow the nullified AoE into the wielder's loadout for 2 turns).
<code>bankDamage</code>	Bank damage taken, release it later (<code>stored_retribution</code>).
<code>enrichSurviveLethal</code>	Survive an otherwise-lethal hit ("One More Breath").

13.2 Passive counters (stateful, accumulate then discharge)

Six stateful counters (`PassiveCounterKind`) accumulate stacks from opponent behavior and then fire. Thresholds from `PassiveCounterConfig`:

Counter	Builds on	Threshold	Payoff
Draegor	Each ability the holder uses (Enmity)	5 → Regret (2t)	If an enemy FAT-chains 2+ abilities while Regret is up, the squad's TEG rises 2 tiers for 2 turns (or, if already SS+, doubles the highest-Affinity ally's Trion Affinity). Max 3/battle.
Nullhymn	Enemy Black Trigger use / status on holder (Discord)	5	Discharges (2×/battle): permanently drops the most-recently-active enemy Black Trigger one resonance grade — or, if none, purges & reflects your team's debuffs.
Reckoning	Enemy crits / 2+ cd abilities vs your team (Debt)	6	Worst offender's cooldowns extended, their next attack forced to a critical miss, their Trion levied.
Gravehour	End of enemy turn, if they dealt no damage or left an enemy ≤30% HP	— (3t cd)	Free uncounterable finisher (+40 flat) on the lowest-HP enemy, who can't be healed next turn.
Coldread	Secretly marks an enemy each turn	— (1t cd)	Correct read alternates a reward (Levy their costliest Trion, then Seize: +2 to the whole squad's rolls for 1 turn). A wrong read docks your next Trion gain.
Ironvow	A random attack type is sanctioned each turn	—	Attack with the sanctioned type for a Sanctioned Strike (unblockable, strips a buff, brands the target). Cost: allies left vulnerable to that type. Max 3/battle.

14. Unique abilities

17 unique abilities (`UniqueBehavior` , excluding the UI-only `clickable`) give the Unique-type characters and select Triggers signature mechanics not expressible as plain damage + status. Key ones (constants in Appendix D):

Unique	Behavior
<code>sharedAgony</code>	Links two targets; damage to one bleeds to the other (×1.2).
<code>graveBargain</code>	Spend 25% of your own HP to power a strike.
<code>martyrsEnd</code>	Below 25% HP, detonate for 80 damage.
<code>vowOfTheDuel</code>	Bind a target: ×2 damage between the two, plus a 2-turn stun.
<code>sunderArms</code>	Heavy single-target armor sunder.
<code>curvingShot</code>	A shot that ignores certain cover/defense.
<code>calledShot</code>	Zero one of the target's stats for a duration.
<code>mindsEye</code>	Reveal an enemy's full loadout (clickable in their panel) for 3 turns.
<code>forcedChoice</code>	Force the target into a lose-lose choice.
<code>memoryTheft</code>	Steal / copy an ability.

Unique	Behavior
sensorySwap	Swap a sense/targeting relationship on 2 targets.
dreadResonance	Damage scales with cumulative damage the target has taken ($\times 0.15$, min 5).
isolation	Cut a target off from ally support.
illusoryDouble	Start with 1 charge; gains charges on ally death; a decoy body.
echoingDoubt	A doubt that backlashes for 20 if acted against.
karmicBind	Two-way link scaling with Team Spirit (25%–60% fractions).
unmaking	A powerful unravelling debuff.

15. Combo recognition

The engine runs a **combo recognizer** over a per-turn action ledger: it detects setup → payoff patterns and rewards them (via TEG Effect 4's payoff advantage and Effect 3's setup → payoff Trion refund). There are **18 recognized combos** across two layers:

- **Layer 1 — generic patterns** (fire on structural conditions): *Control then Strike, Affliction Detonate, Setup Exploit, Focus Fire.*
- **Layer 2 — signature chains** (specific trigger/status pairings): *Frozen Detonation, Terror Cascade, Shatterpoint Execution, Piercing Corrosion, Pinned Shot, Mind Unravel, Charmed Opening, Flashfire, Opening Sweep, Venom Reap, Snared Execution, Brittle Pierce, Blinding Volley, Dread Shatter.*

The recognizer is fed live during resolution and cleared each turn. A design-time proposer (`tool/propose_signature_combos.dart`) helps author new signature chains.

16. Team Efficiency Grade (TEG)

Before battle, each squad's Loadout is scored into a **Team Efficiency Grade** — a build-quality meta-stat that both **helps you in battle** and **inversely gates your XP**, making the Grade a genuine risk/reward dial rather than a strictly better number.

Six sub-scores (weighted → composite 0–100 → tier):

Sub-score	Weight	Measures
Team Spirit Alignment	25%	Whether each character's loadout leans offense/sustain matching their Team Spirit pole.
Stat Coherence	—	Whether stats reinforce the intended role.
Loadout Synergy	—	How well the equipped Triggers combo.
Perk Utilization	—	Whether the loadout enables the character's perk.
Trion Economy	—	Whether the build's costs fit its income.
Resonance Fit	—	Black Trigger fit (null if no Black Trigger).

Tiers: D, C, B, A, S, SS, SSS.

In-battle effect (TEG Effects 1–5): the Grade feeds a per-team **dice advantage** into the to-hit, defense, and resistance rolls (coordination bonus + an inverted "Operator's Read" on defense), an **SSS crit-range widener**, a **recognized-combo payoff advantage** (Effect 4), and a **setup → payoff Trion refund** (Effect 3). Draegor's counter can shift a squad's effective Grade up 2 tiers.

Post-battle XP (the inverse counterweight): a weaker Grade earns *more* XP, an elite Grade pays an "elite tax":

Tier	D	C	B	A	S	SS	SSS
XP bonus	+75%	+63%	+51%	+40%	+28%	+16%	+5%

Awarded XP = `round(baseXp × (1 + bonus%))` , computed **server-side** (§20).

17. Black Triggers

Black Triggers are rare, powerful loadout pieces — effectively a whole extra kit bolted onto one operator (a suite of active abilities, passives, and sometimes a once-per-battle "World ability"). There are **9**:

Black Trigger	Identity
Ashbringer	High-damage melee/ranged pair on short cooldowns.
Fracture Edge	A burst attack plus offense-focused passives.
Solar Flare	A single massive AoE nuke on a long cooldown, plus a Critical passive.
Aegis Core	A World ability that negates the first few instances of damage.
Bastion Frame	Armor, Damage Resistance, Status Resistance, and Foresight passives.
Wellspring	A direct heal, a Regenerating buff, and a Mirror Ward reflect.
Chorus Bond	Team Spirit and healing-focused passives — a support suite.
Paradox Shard	An unpredictable debuff ability plus a disruptive passive.
Gravebind	A World ability: once per battle, survive a hit that would defeat you.

A Black Trigger's fit is scored by the TEG **Resonance Fit** sub-score, and the **Nullhymn** counter can permanently downgrade an enemy Black Trigger's resonance.

18. Character perks

Each character has one **innate perk**, always active, that a good loadout should enable (the TEG Perk Utilization sub-score rewards this):

Character (type)	Perk	Effect
Kaito Reyes (atk)	Last Ace	Critical Chance doubled while Kaito is the last living teammate.
Vela Ashworth (atk)	First Blood	First attack of the battle gains bonus Critical Chance.
Dross (atk)	Overwhelm	AoE abilities deal bonus damage to a target that already has a debuff.
Ren Kobayashi (atk)	First Strike	Flat Attack bonus on the first ability used each battle.
Airi Tanaka (atk)	Feint	The first attack against Airi each battle is rolled with disadvantage.
Marren Osei (def)	Bulwark	Armor doubled while any living teammate remains.
Ilona Vance (def)	Riposte	When a melee attack against her misses, she gains a stacking Attack buff.
Bastian Cole (def)	Absorb	First instance of damage each battle is halved.
Dorian Voss (def)	Immovable	Resists forced movement / displacement effects.
Sable Whitlock (def)	(defensive anchor)	High Defense/Armor stat profile.
Priya Nakamura (sup)	Combat Medic	Heal-over-time effects heal for more.
Soren Talvik (sup)	Weaken Resolve	Debuffs hit harder vs a target already carrying one of his debuffs.
Yuki Amaral (sup)	Devoted Aid	+10% to all healing, any source.
Haru Ellison (sup)	Battery	Trion battery — very high Trion Affinity (32) fuels the squad.
Celestine Moreau (sup)	Warding Presence	Ally-targeted buffs also grant a resistance bonus.

Character (type)	Perk	Effect
Zheng Anders (unq)	Foresight	Once per battle, reroll one of his own attack rolls.
Nadia Kessler (unq)	Chain Reaction	Each additional ability in a FAT turn deals more than the last.
Rurik Voss (unq)	All or Nothing	Glass-cannon profile (Attack 29 / Defense 6): high risk, high reward.
Mireille Song (unq)	Decoy	Once per battle, an attack against her may simply miss.
Tobias Renner (unq)	Versatile	Small bonus to whichever stat matches the last ability category used.

19. AI opponents

The opponent is an **AI profile** (skill classes: Beginner, Amateur, Intermediate, Professional, Expert). The AI **builds and resolves a full queue through the same resolution path as the player**, with a simulated "thinking" delay before it commits, so single-player and future server-authoritative play share one code path. AI tuning (valuing the counters/uniques/statuses well) is a tracked future phase.

20. Accounts & XP (server-authoritative backend)

Accounts and XP are **live on a Supabase backend** (project `zszjkanssxhejhotbrca`), behind the `AccountService` / `XpLedger` interfaces via a `Services` locator that falls back to local stubs if the backend is unreachable.

- **Stress-free auth:** guest (anonymous), passwordless email (magic link), and Google OAuth; upgrading a guest keeps the same id (no lost progress).
- **Server-authoritative XP:** the client reports a battle result; the Postgres function `award_battle_xp` **recomputes the inverse-TEG multiplier server-side** and owns the running total, so a client can never set its own XP. Row Level Security restricts every player to their own rows.
- A daily **keep-alive** GitHub Action prevents the free-tier project from pausing.

Setup and schema are documented in `docs/supabase_setup.md` and `supabase/schema.sql`.

21. Build status

- **Engine (Phases A–E, I, J):** done & merged — queue engine, reactive + passive counters, unique subtype + 17 uniques, 20/20/20-ish trigger rebalance, combo recognition, TEG mechanical effects + inverse-TEG XP.
- **Accounts / XP (§20):** live and verified end-to-end.
- **Phase F UI:** TEG badge, hidden stats, Team Spirit readout, loadout-builder pass, passive-counter descriptions, clickable-portrait detail panel + Mind's Eye reveal, sign-in flow, post-battle XP readout, and the battle-log rework (tap-to-expand, plain-English breakdowns, clickable entity popups). Remaining: queue display + resolve-beat polish.
- **Balance:** the 5 near-duplicate trigger clusters have been differentiated; a full balance pass (bounded accuracy, initiative, status/trigger budgets) is proposed in `docs/reviews/` and pending.

Appendix A: Full Trigger catalog (60 active Triggers)

Trion = use cost · cd = cooldown (turns) · eq = equip cost · sub = subtype · h = hits/use · tc = targets · special = unique behavior or reactive kind.

Attacker (16)

Trigger	Trion	cd	eq	sub	h	tc	damage	status	special
"Martyr's End"	10	4	28	unique	1	3	—	—	martyrsEnd
Cinderburst	18	2	26	aoe	1	3	1d6+18	scorched	—
Cleave	15	1	24	aoe	1	2	1d6+18	shattered_guard	—
Cryo Burst	18	2	26	aoe	1	3	1d6+14	chilled	—
Dread Resonance	18	2	26	unique	1	1	—	—	dreadResonance
Dread Wave	18	2	26	aoe	1	3	1d6+12	overwhelmed	—
Frost Lance	14	1	22	single	1	1	1d6+18	chilled	—
Grave Bargain	8	2	14	unique	1	1	—	—	graveBargain
Piercing Thrust	14	1	22	single	1	1	2d6+45	—	—
Predictive Parry	20	2	20	single	1	1	—	—	dodgeMeleeSingle
Shared Agony	12	2	24	unique	1	1	2d6+40	—	sharedAgony
Soul Siphon	16	2	24	single	1	1	heal 1d4+-1	—	—
Sunder Arms	16	3	28	unique	1	1	2d6+30	—	sunderArms
Twin Fang Strike	10	1	20	single	1	1	2d6+37	—	—
Vow of the Duel	15	3	24	unique	1	1	—	—	vowOfTheDuel
Whirlwind Slash	12	1	20	aoe	1	3	1d4+16	bleeding	—

Shooter (9)

Trigger	Trion	cd	eq	sub	h	tc	damage	status	special
Arc Volley	20	2	26	burst	3	2	1d4+10	—	—
Frag Grenade	18	2	26	aoe	1	3	1d6+16	—	—
Gatling Burst	24	2	28	burst	5	1	1d4+8	exposed	—
Pepper Shot	16	1	14	burst	3	1	1d4+12	—	—
Rapid Fire	18	2	16	burst	3	1	1d4+14	bleeding	—
Scattershot	20	2	26	burst	4	3	1d4+6	slowed	—
Split Shot	18	2	16	burst	2	2	1d6+10	—	—
Suppressing Fire	20	2	26	burst	3	2	1d4+16	suppressed	—
Thunderclap Round	18	2	26	aoe	1	3	1d4+10	overwhelmed	—

Sniper (3)

Trigger	Trion	cd	eq	sub	h	tc	damage	status	special
Called Shot	16	2	24	unique	1	1	—	—	calledShot
Curving Shot	18	2	26	unique	1	1	2d6+30	—	curvingShot
Longshot	30	2	34	single	1	1	4d8+86	—	—

Trapper (24)

Trigger	Trion	cd	eq	sub	h	tc	damage	status	special
Acid Spray	16	2	24	aoe	1	3	1d4+10	corroded	—
Caustic Cloud	16	2	16	aoe	1	3	1d4+6	poisoned	—
Charm Whisper	20	2	26	single	1	1	—	charmed	—
Death Ledger	18	2	20	single	1	1	1d4+8	—	nullifyAoe

Trigger	Trion	cd	eq	sub	h	tc	damage	status	special
Dread Gaze	14	1	20	single	1	1	1d4+8	terrified	—
Echoing Doubt	16	3	24	unique	1	1	—	—	echoingDoubt
Flashbang Round	18	2	22	aoe	1	3	—	blinded	—
Forced Choice	14	2	22	unique	1	1	—	—	forcedChoice
Isolation	14	3	22	unique	1	1	—	—	isolation
Karmic Bind	16	3	26	unique	1	1	—	—	karmicBind
Mass Confusion	20	2	26	aoe	1	3	1d4+8	silenced	—
Memory Theft	16	3	24	unique	1	1	—	—	memoryTheft
Mind Fog	16	2	14	aoe	1	3	1d4+8	blinded	—
Mind Shatter	18	2	24	single	1	1	1d4+8	silenced	—
Nightmare Pulse	18	2	26	aoe	1	2	1d6+10	terrified	—
Numbing Toxin	20	2	18	single	1	1	—	—	burstMitigation
Psychic Scream	18	2	26	aoe	1	3	1d4+8	silenced	—
Root Snare	18	2	20	single	1	1	1d4+8	forced_repetition	—
Scramble	20	2	24	single	1	1	1d4+8	misfire	—
Sensory Swap	14	2	22	unique	1	2	—	—	sensorySwap
Shatterpoint	12	1	18	single	1	1	1d6+18	corroded	—
Unmaking	18	3	26	unique	1	1	—	—	unmaking
Venom Needle	12	1	18	single	1	1	1d4+12	poisoned	—
Venom Spray	20	2	24	burst	3	2	1d4+6	poisoned	—

Optional (8)

Trigger	Trion	cd	eq	sub	h	tc	damage	status	special
"Guardian's Aegis"	14	2	18	single	1	1	—	guarded, braced	—
"Mind's Eye"	10	3	20	unique	1	1	—	—	mindsEye
Cleansing Ward	16	2	22	single	1	1	—	regenerating, warded	—
Frozen Tempo	18	2	20	single	1	1	—	—	cooldownSabotage
Illusory Double	0	2	24	unique	1	1	—	—	illusoryDouble
Rally Cry	14	2	18	aoe	1	3	—	inspired	—
Stored Retribution	16	2	18	single	1	1	—	—	bankDamage
War Chant	10	2	14	single	1	1	—	empowered	—

Appendix B: Character roster (20)

Character	Type	ATK	DEF	ARM	HP	Trion	T.Aff	T.Spirit	Crit	FAT	Perk
Kaito Reyes	attack	27	8	1	100	110	20	35	15	12	Last Ace
Vela Ashworth	attack	26	7	1	100	105	18	40	12	10	First Blood
Dross	attack	24	9	2	100	115	16	35	6	9	Overwhelm
Ren Kobayashi	attack	26	8	1	100	100	22	30	10	14	First Strike
Airi Tanaka	attack	22	9	1	100	100	20	45	13	11	Feint
Marren Osei	defense	12	16	3	100	120	16	55	3	7	Bulwark
Ilona Vance	defense	17	15	2	100	110	18	50	5	8	Riposte

Character	Type	ATK	DEF	ARM	HP	Trion	T.Aff	T.Spirit	Crit	FAT	Perk
Bastian Cole	defense	14	14	3	100	115	16	50	3	7	Absorb
Dorian Voss	defense	15	14	2	100	110	18	48	4	8	Immovable
Sable Whitlock	defense	14	15	3	100	115	18	55	4	8	—
Priya Nakamura	support	10	10	1	100	105	24	65	4	9	Combat Medic
Soren Talvik	support	12	9	1	100	105	20	60	5	9	Weaken Resolve
Yuki Amaral	support	10	11	2	100	105	22	70	4	8	Devoted Aid
Haru Ellison	support	10	10	1	100	130	32	60	4	9	Battery
Celestine Moreau	support	10	11	2	100	110	20	62	4	8	Warding Presence
Zheng Anders	unique	19	10	1	100	105	24	45	8	12	Foresight
Nadia Kessler	unique	22	9	1	100	100	20	40	9	16	Chain Reaction
Rurik Voss	unique	29	6	1	100	100	18	30	14	13	All or Nothing
Mireille Song	unique	19	9	1	100	100	20	45	8	11	Decoy
Tobias Renner	unique	19	10	1	100	105	20	45	7	11	Versatile

Appendix C: Status effect magnitudes (selected)

Constant	Value
bleedingDamagePerTurn	8
scorchedDamagePerTurn	12
necroticWoundDamagePerTurn	12
poisonedDurationTurns	3
acidArmorReduction	5
corrodedArmorReduction	3
markedAllDamageTakenMultiplier	1.5
exposedAllDamageTakenMultiplier	1.25
guardedAllDamageTakenMultiplier	0.75
petrifiedAllDamageTakenMultiplier	0.5
empoweredOutgoingDamageMultiplier	1.25
enragedOutgoingDamageMultiplier	1.5
weakenedOutgoingDamageMultiplier	0.75
stunnedDurationTurns	1
frozenDurationTurns	1
charmedDurationTurns	3
silencedDurationTurns	1
sappedDrainPercentOfTrionCapacity	0.25
genjutsuTrappedDrainPercentOfTrionCapacity	0.15
adrenalineRushCriticalChanceBonus	15
battleTranceFatChanceBonus	20
ralliedMaxHealthBonus	20
inspiredAttackBonus	2
fatiguedAttackPenalty	2

Appendix D: Tunable constants

Trion income

Constant	Value
baseChanceLowToMedium	0.35
baseChanceMediumToHigh	0.20
affinityWeightPerPoint	0.01
lowAmount	10
mediumAmount	20
highAmount	35

FAT

Constant	Value
baseFatCooldownTurns	3
maxAbilitiesOnFatTrigger	3
normalAbilitiesPerTurn	1
multiAbilityPenaltyThreshold	2
cooldownDoubleMultiplier	2.0
trionAffinityPenaltyMultiplier	0.5

Combat / crit

Constant	Value
criticalHitDamageMultiplier	2.0
criticalMissDefensePenaltyPct	0.20
criticalMissTeamSpiritPenaltyPct	0.20
thresholdAtMinChance	20
thresholdAtMaxChance	5
minChancePercent	0
maxChancePercent	90

Loadout & Team Spirit

Constant	Value
maxEquippedTriggers	8
requiredActiveAbilityCount	4
statMin	0
statMax	100
midpoint	50
maxSingleTargetDamageBonus	0.30
maxBurstDamageBonus	0.30
maxCriticalChanceBonus	20
maxHealthRegenBonus	0.30
maxFatChanceBonus	20
secondsPerTurn	15))

Passive counters

Constant	Value
draegorEnmityThreshold	5
draegorMaxRegretPerBattle	3
nullhymnDiscordThreshold	5
nullhymnMaxDischarges	2
reckoningDebtThreshold	6
gravehourCooldownTurns	3
gravehourFinisherFlatDamage	40
gravehourLowHpThreshold	0.3
coldreadSeizeRollBonus	2
ironvowMaxSanctionedStrikes	3

Unique abilities

Constant	Value
sharedAgonyLinkedDamageMultiplier	1.2
graveBargainHpSpendFraction	0.25
martyrsEndHpThreshold	0.25
martyrsEndDamage	80
dreadResonanceDamagePerCumulativeDamage	0.15
dreadResonanceMinDamage	5
illusoryDoubleStartingCharges	1
karmicBindLowTsFraction	0.25
karmicBindHighTsFraction	0.60
echoingDoubtBacklashDamage	20
vowOfTheDuelDamageMultiplier	2.0
vowOfTheDuelStunDurationTurns	2

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